

MUHAMMAD ARGYA VITYASY

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EDUCATION

Universitas Gadjah Mada

Yogyakarta, Indonesia

Bachelor of Computer Science | Cumulative GPA: 3.85/4.00 | Expected Graduation: 2027

August 2023 – Present

- **Relevant Coursework:** Algorithms and Data Structures, Scalable Software Engineering, Software Engineering Methods, Cloud Computing, Computer Networks, Cyber Security

WORK EXPERIENCE

Software Engineer Intern | GDP Labs

February 2026 – June 2026

Enterprise AI automation platform building multi-agent LLM systems for HR, payroll, and project management workflows.

- Built **LLM-powered agents** with structured outputs and tool execution, orchestrating multi-step HR and project management workflows across 3+ workflow types via routing and provider fallback.
- Built a **compliance monitoring pipeline** with scheduled event triggers and automated audit report generation, replacing manual report tracking across organizational workflows.
- Built a **multi-tenant IAM SDK** standardizing authentication across enterprise projects, handling tenant isolation and provider synchronization with **atomic multi-provider transactions** to prevent partial-state failures on rollback.
- Authored **Playwright end-to-end test suites** integrated into CI to catch regressions and validate multi-agent tool execution across workflow paths before production deployment.

Backend Engineer | PT Satria Duta Raya

October 2025 – January 2026

B2B procurement platform enabling businesses to discover ranked products through an asynchronous recommendation pipeline.

- Architected a **Go/Gin backend** with 6 domain boundaries isolating business logic to reduce coupling and simplify feature ownership across a B2B procurement platform.
- Built an **event-driven product ranking pipeline** on **Redis Streams** with idempotent consumers, Dead Letter Queues, and retry backoff, decoupling heavy computation from request paths and containing failure blast radius.
- Shipped **full-stack observability** using **Prometheus and Grafana** with dashboards and alerting on background worker failures, reducing time to diagnose and resolve production incidents.

Backend Engineer | Jogja International Model United Nation 2025

April 2025 – October 2025

Annual academic simulation of the United Nations bringing together students worldwide for diplomatic debates.

- Maintained stable service during **500+ concurrent registration spikes** by optimizing database connection pooling and query performance, deployed via **Docker** with Go and PostgreSQL.
- **Cut manual staff workload by 40%** by building a **role-based admin dashboard** with an approval pipeline that automated user verification and permission-managed workflows for 150+ staff actions.

Backend Engineer | Organisasi Mahasiswa Ahli Teknologi Informasi UGM

March 2024 – July 2025

Student-led IT organization comprising 8 specialized divisions and 80+ members managing real-world software projects.

- Built a **reusable Go/Node.js/Docker backend framework** deployed across 4 client projects, **cutting development and deployment time by 60%**.
- Handled **3,000+ user registrations** and high-concurrency workloads with stable uptime across all deployed projects.
- Led weekly technical discussions for 15 engineers, driving adoption of **Docker-based deployment standardization** and shared backend tooling across project teams.

PROJECTS

National University Entrance Exam Mock Platform | [GitHub](#)

- Served **200+ concurrent users** at **sub-50ms latency** across a 4-service **Go/Docker microservices architecture** with token-based auth and per-service database isolation.
- Processed **150+ timed exam sessions** from a 1,000-question bank with **<5% error rate** via a custom exam engine exposing 20+ REST APIs and connection-pooled PostgreSQL queries.

ADDITIONAL SKILLS & INTERESTS

Languages & Frameworks: Go, Python, TypeScript, Gin-Gonic, Express, Pydantic

Data & Infrastructure: PostgreSQL, MongoDB, Redis, Kafka, Docker, Colima, Terraform, Ansible, AWS S3, MinIO

Observability & Testing: Prometheus, Grafana, Playwright, OpenTelemetry

Interests: LLM-powered automation, distributed systems, cloud computing